

UNDERSTANDING THE FINE-GRAINED KNOWLEDGE CAPABILITIES OF VISION-LANGUAGE MODELS

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ABSTRACT

Vision-language models (VLMs) have achieved strong performance on visual reasoning, document understanding, and multimodal dialogue benchmarks across diverse base models, alignment architectures, and training corpora. These benchmark gains, however, do not by themselves establish fine-grained visual knowledge: the capacity to distinguish visually similar categories that is central to many recognition-intensive uses of VLMs. We evaluate a broad set of recent VLMs on four fine-grained classification benchmarks and compare these results with standard VLM evaluation. The analysis shows that fine-grained recognition is a distinct dimension of visual capability, with substantial variation among models that obtain similar general VQA scores and persistent gaps between most VLMs and their underlying CLIP-style vision encoders. Through 22 systematic ablations, we further isolate the design choices that govern this capability. Stronger LLMs improve fine-grained and general benchmark performance at comparable rates, whereas stronger vision encoders yield disproportionately larger gains on fine-grained classification once they are integrated through pretraining. Pretraining is also a central determinant of fine-grained performance, especially when language-model weights are unfrozen after connector warmup. These findings identify the components that most directly preserve and enhance fine-grained visual knowledge in VLMs and provide concrete evidence for developing more vision-centric multimodal systems.

1 INTRODUCTION

Recent vision-language models (VLMs), which couple visual encoders with large language models (LLMs), now support increasingly sophisticated forms of visual interpretation and multimodal reasoning (Alayrac et al., 2022; Team et al., 2023; OpenAI, 2023; Liu et al., 2023; Wang et al., 2024b). Their progress is especially visible on open-ended visual question answering, document understanding, and multimodal dialogue benchmarks (Goyal et al., 2017; Yue et al., 2024; Mathew et al., 2021; OpenAI, 2023). Yet this success leaves unresolved a more vision-centric question: how much fine-grained visual knowledge do contemporary VLMs retain and use, and which architectural or training choices most effectively improve that capability?

Fine-grained visual recognition—the ability to distinguish among visually similar categories—requires representations that encode subtle discriminative evidence while remaining robust to intra-class variation in pose, illumination, viewpoint, and background. Standalone vision encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) perform strongly on fine-grained classification benchmarks, but the corresponding behavior of newer LLM-based VLMs remains less clearly characterized. This gap matters because the multimodal interface may either preserve the fine-grained visual knowledge present in the encoder or attenuate it during alignment with language.

Fine-grained visual knowledge is not a peripheral classification skill; it is part of the perceptual substrate on which reliable downstream reasoning depends. Many practical VLM settings involve long-tailed category distributions in which closely related labels imply materially different decisions. A model that misidentifies a mushroom species, for example, cannot reliably use textual knowledge to infer toxicity or determine whether the mushroom is safe to consume. Measuring and explaining fine-grained knowledge acquisition therefore informs both the evaluation of existing VLMs and the design of future architectures and training pipelines.

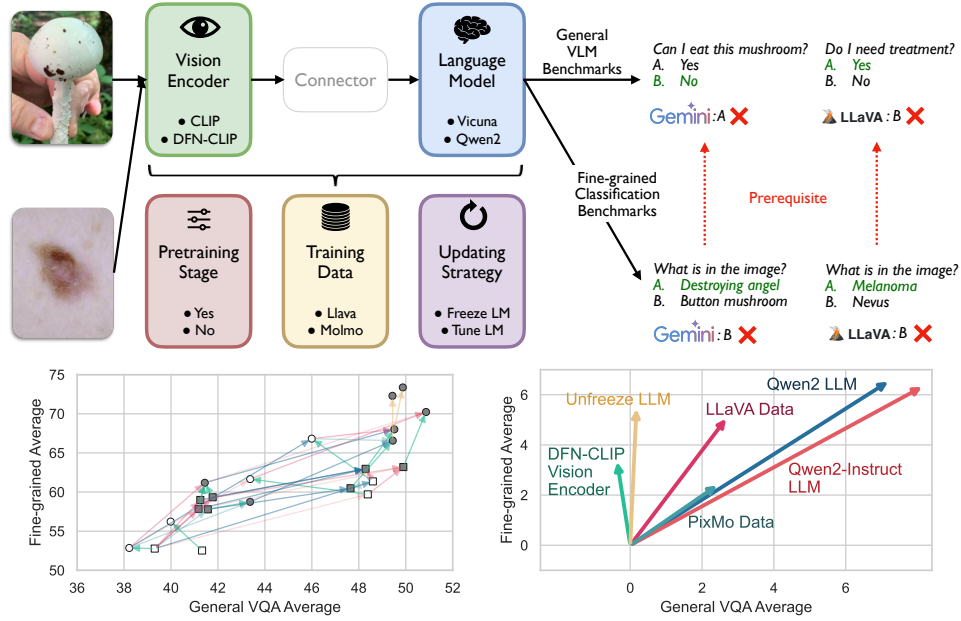


Figure 1: Overview. (Top) We evaluate fine-grained classification in vision-language models (VLMs), a vision-centric capability that underlies higher-level understanding and reasoning but is only indirectly reflected in many standard VLM benchmarks. (Bottom) Across 22 controlled ablations of model components and training strategies (left), we identify the factors that most strongly determine fine-grained classification performance in VLMs (right).

We first present a comprehensive evaluation of fine-grained visual knowledge in state-of-the-art VLMs by comparing performance on fine-grained classification with performance on general VLM tasks. Specifically, we evaluate 15 VLMs, including LLaVA (Liu et al., 2023; 2024a), Phi (Abdin et al., 2024), Qwen2-VL (Wang et al., 2024b), and Molmo (Deitke et al., 2024), on ImageNet (Deng et al., 2009), Flowers (Nilsback & Zisserman, 2008), Pets (Parkhi et al., 2012), and Food (Bossard et al., 2014). We contrast these results with aggregated scores from eight general VLM benchmarks, including MMMU (Yue et al., 2024), MathVista (Lu et al., 2024), and MMVet (Yu et al., 2024). The comparison establishes that models with similar general VLM scores can differ sharply in fine-grained classification accuracy, making fine-grained recognition a distinct axis of visual capability that standard VLM benchmarks do not adequately capture. We also find a persistent gap between VLMs and their corresponding CLIP-based vision encoders, consistent with prior evidence that multimodal alignment can fail to preserve all recognition capability available in the visual backbone (Zhang et al., 2024; Geigle et al., 2024).

To explain these empirical patterns, we conduct a systematic ablation study of the model components and training choices that govern fine-grained classification performance. The study isolates the effects of LLM choice, vision-encoder choice, pretraining, parameter-updating strategy, and data quality within a common LLaVA-style evaluation framework. Across 22 ablation experiments, the paper makes the following contributions:

- **Language model (§4.2):** Stronger base language models improve fine-grained classification and general VLM benchmarks at comparable rates, indicating that language-model quality benefits both evaluation regimes without specifically targeting fine-grained recognition.
- **Vision encoder (§4.2):** Stronger vision encoders produce larger gains on fine-grained classification than on general VLM benchmarks, particularly after two-stage training integrates the visual representation into the multimodal model.
- **Pretraining stage (§4.3):** Image-caption pretraining substantially improves fine-grained classification and yields smaller gains on general VLM benchmarks, identifying pretraining as a primary mechanism for transferring visual recognition capacity into the VLM.
- **Weight updating method (§4.3):** Updating both the vision-language connector and the LLM during pretraining improves fine-grained classification relative to connector-only pre-

training, showing that the parameter-updating strategy affects how effectively visual knowledge is absorbed.

- **Data quality (§4.3):** Within the tested settings, pretraining and instruction-tuning data quality has limited impact on fine-grained classification, especially when language-model weights remain frozen during pretraining.

Together, these results reposition fine-grained visual perception as a necessary component of VLM evaluation rather than an auxiliary classification task. By connecting benchmark discrepancies to specific modeling and training choices, our study clarifies how VLMs acquire, preserve, or lose fine-grained visual knowledge and provides evidence-backed guidance for building more vision-centric models for recognition-intensive and long-tail settings.

2 RELATED WORK

Vision-language models and their evaluation. Vision-language models (VLMs) integrate visual inputs with language-model reasoning, most commonly through an LLaVA-style architecture (Liu et al., 2023) consisting of a vision encoder, a language model, and an MLP connector that maps visual representations into the language-model interface. This design family has produced strong results across multimodal tasks, including visual question answering (VQA), document understanding, and general reasoning (Liu et al., 2023; Dai et al., 2023; Awadalla et al., 2023; Wang et al., 2024b). Correspondingly, VLM evaluation has increasingly centered on benchmarks such as MMMU (Yue et al., 2024), MathVista (Wang et al., 2024a), and DocVQA (Mathew et al., 2021). These benchmarks are valuable for measuring reasoning and language use conditioned on visual inputs, but they do not directly characterize core recognition abilities such as object identification and fine-grained classification. Our work addresses this gap by evaluating fine-grained classification as a vision-centric capability and by analyzing which VLM design choices determine performance on that axis.

Fine-grained visual classification. Fine-grained classification is a long-standing computer vision task concerned with discriminating visually similar subcategories within broader object classes, including bird species (Wah et al., 2011), flowers (Nilsback & Zisserman, 2008), and pet breeds (Parkhi et al., 2012). Such benchmarks have been central to the development and evaluation of vision encoders, including ResNet (He et al., 2016), CLIP (Radford et al., 2021), and DINO (Caron et al., 2021). Despite this role in vision research, recent VLM evaluation protocols often prioritize VQA-style reasoning benchmarks over recognition benchmarks that expose fine-grained visual knowledge. Because long-tail applications frequently require precise distinctions among visually similar categories, fine-grained classification provides an essential complement to language- and reasoning-centric VLM evaluation.

Evaluation and ablation of VLMs for fine-grained classification. Recent studies have begun to test VLMs on fine-grained classification (Zhang et al., 2024; Geigle et al., 2024), but existing analyses are often limited to a small number of models, datasets, or evaluation conditions. We extend this line of work by evaluating a broader set of VLMs across multiple fine-grained classification benchmarks under a unified multiple-choice protocol. More importantly, we move beyond measurement alone: through controlled ablations, we identify the model components and training decisions that explain differences in fine-grained performance.

Several recent works examine the broader VLM design space, including architecture choices (McKinzie et al., 2024), training strategies (Karamcheti et al., 2024), and multimodal data composition (Gadre et al., 2023; Udandarao et al., 2024; Deitke et al., 2024). These studies primarily optimize or analyze performance on general benchmarks that emphasize language-mediated reasoning. Our study instead asks how base-model selection, pretraining procedure, and pretraining or instruction-tuning data affect fine-grained classification, thereby isolating design principles for VLMs whose visual recognition capability must remain strong after multimodal alignment.

3 EVALUATION: FINE-GRAINED CLASSIFICATION

This section motivates fine-grained classification as a VLM evaluation target, specifies the benchmark conversion and testing protocol, and reports the empirical patterns observed across 15 recent VLMs.

3.1 MOTIVATION

Fine-grained classification is a foundational computer vision task that requires distinguishing visually similar subcategories within a broader category, such as bird species or car models (Wei et al., 2021). Benchmarks such as ImageNet (Deng et al., 2009) and Flowers (Nilsback & Zisserman, 2008) have therefore served as standard testbeds for evaluating whether vision encoders capture discriminative visual structure. With large-scale self-supervised and supervised learning, encoders such as CLIP (Radford et al., 2021) and DINO (Caron et al., 2021) have reached near- or superhuman performance on challenging fine-grained classification tasks.

The same benchmark family has received much less attention in the evaluation of vision-language models (VLMs). Although recent VLMs such as LLaVA (Liu et al., 2023) show strong ability to interpret visual inputs, they are typically assessed through VQA benchmarks such as MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024), or through document-oriented tasks such as DocVQA (Mathew et al., 2021) and TextVQA (Singh et al., 2019). These evaluations measure important reasoning and language-processing capabilities, but they contain limited coverage of fine-grained object categories and thus leave a central recognition capability underexamined.

Fine-grained classification is especially important for VLMs because many real-world inputs follow long-tailed distributions in which adjacent categories require different actions or interpretations. If a blind user asks whether a mushroom is safe to eat, for example, the model must identify the species before applying textual knowledge about toxicity; confusing a deadly destroying angel with an edible white button mushroom could have severe consequences. Analogous risks arise in medical diagnosis, where visually similar conditions may require different treatments, and in autonomous driving, where visually similar traffic signs can demand different maneuvers. These examples motivate a direct evaluation of whether VLMs preserve the recognition precision needed to support reliable downstream reasoning.

3.2 BENCHMARKS

We evaluate fine-grained visual knowledge using four established object-recognition benchmarks that cover complementary recognition domains: **ImageNet-1K** (Deng et al., 2009), **Oxford Flowers-102** (Nilsback & Zisserman, 2008), **Oxford-IIIT Pet-37** (Parkhi et al., 2012), and **Food-101** (Bossard et al., 2014).

Because these benchmarks contain 37 to 1,000 classes, direct evaluation in the multiple-choice format commonly used for VLM training and testing is nontrivial. We therefore follow Geigle et al. (2024) and convert each dataset into 5-way multiple-choice questions. For each image, an OpenCLIP ViT-L/14 model (Schuhmann et al., 2022) selects the four hardest negative labels according to image-label cosine similarity, and these labels are combined with the ground-truth class to form the answer set. We use OpenCLIP to avoid systematically biasing the negative choices against VLMs built on OpenAI-CLIP (Radford et al., 2021), SigLIP (Zhai et al., 2023), or DFN-CLIP (Fang et al., 2023), whose training data differ from the OpenCLIP model. This conversion preserves much of the classification difficulty while placing all models in a familiar multiple-choice interface. For example, CLIP ViT-L/14-336px (Radford et al., 2021) obtains 75.2% accuracy on the original 102-class Flowers-102 benchmark and 78.3% accuracy on our 5-way multiple-choice version.

3.3 TESTING SETUP

We evaluate 15 recent VLMs in the 7B–13B parameter range on the converted 5-choice fine-grained classification benchmarks (Liu et al., 2023; 2024a; Chen et al., 2024a; Abdin et al., 2024; Deitke et al., 2024; Laurençon et al., 2024; Hong et al., 2024; Chen et al., 2024c; Wang et al., 2024b). For each VLM, we use the corresponding default prompt format from VLMEvalKit (Duan et al., 2024) and score accuracy by exact match to the multiple-choice options. For baseline comparison, we evaluate CLIP models (Radford et al., 2021; Schuhmann et al., 2022; Fang et al., 2023) with the standard zero-shot protocol: images and class labels formatted as “A photo of a [class]” are encoded by the CLIP image and text encoders, and the top-1 prediction is the label whose text embedding has the highest cosine similarity to the image embedding.

To test whether fine-grained classification is captured by existing VLM evaluation, we compare average fine-grained accuracy across the four classification benchmarks with each model’s average

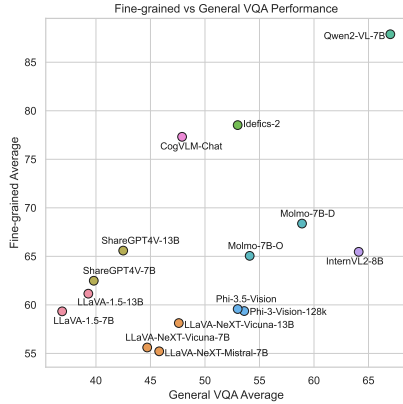


Figure 2: **Fine-grained classification compared with general VQA performance across VLMs.** Recent VLMs with similar aggregate VQA scores can differ substantially in fine-grained classification accuracy, showing that standard VQA benchmarks do not directly measure this recognition capability.

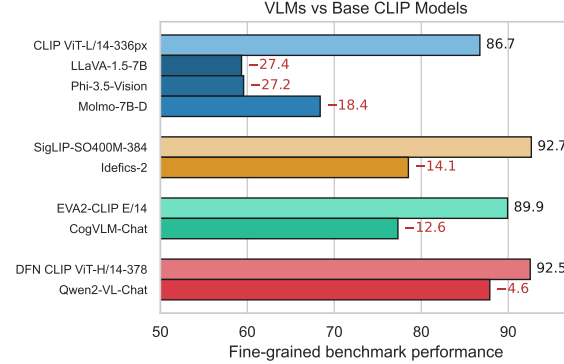


Figure 3: **Fine-grained classification performance of VLMs and their corresponding CLIP vision encoders.** Qwen2-VL-Chat nearly matches its DFN-CLIP vision encoder, whereas the other evaluated VLMs remain substantially below their encoder baselines, indicating considerable headroom in preserving fine-grained visual knowledge during multimodal alignment.

score on eight general VQA benchmarks from VLMEvalKit, including MMMU (Yue et al., 2024) and MathVista (Lu et al., 2024).

3.4 RESULTS

Figure 2 plots average fine-grained classification accuracy against average general VQA performance for all evaluated VLMs. The relationship is not one-to-one: models with similar general VQA scores can exhibit large differences in fine-grained classification accuracy. For instance, CogVLM-Chat (Hong et al., 2024) and LLaVA-NeXT-Vicuna-13B (Liu et al., 2024a) both achieve nearly 48% average accuracy on general VQA benchmarks, yet their fine-grained classification accuracy differs by 19pp, with 77.3% for CogVLM-Chat and 58.1% for LLaVA-NeXT-Vicuna-13B. This discrepancy establishes fine-grained classification as a distinct dimension of VLM visual capability rather than a quantity that can be inferred from standard VQA performance.

Finding 1: Fine-grained classification represents a distinct aspect of visual capability that current VLM benchmarks fail to adequately measure.

Current VLMs are built on pretrained vision encoders, which makes it possible to compare each multimodal model with the recognition performance of its visual backbone under the same multiple-choice formulation. Figure 3 reports this comparison for selected VLMs and their corresponding CLIP-model baselines. The results show a substantial and model-dependent gap: Molmo (Deitke et al., 2024), despite strong performance on several VQA benchmarks, obtains 68.4% average accuracy in our multiple-choice fine-grained evaluation, whereas its CLIP ViT-L/14 (Radford et al., 2021) vision encoder obtains 86.7%. Even Qwen2-VL-Chat (Wang et al., 2024b), the strongest tested VLM in this comparison, remains 4.6 points below its DFN-CLIP (Fang et al., 2023) vision encoder. These gaps show that fine-grained knowledge present in the visual encoder is not automatically preserved after multimodal alignment, motivating the ablation study in the next section.

Finding 2: A significant performance gap exists between VLMs and CLIP models in fine-grained classification tasks.

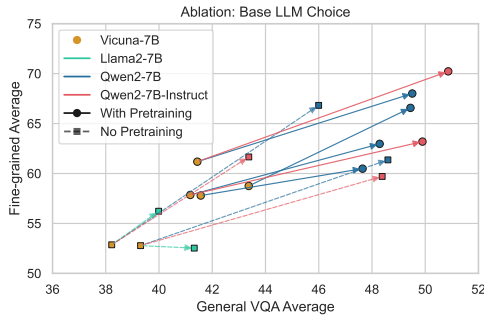


Figure 4: **Ablating base LLM, swapping out Vicuna-7B from LLaVA with other LLMs.** On average, switching from Vicuna to Qwen2-7B results in a +7.5pp increase in fine-grained performance and a +7.7pp improvement in general VQA performance.

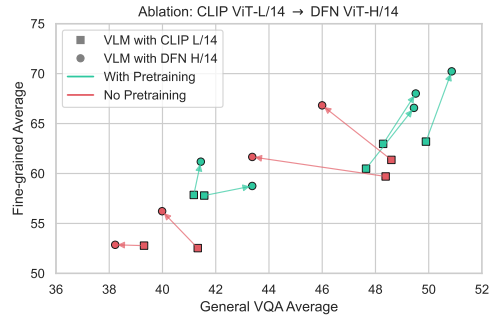


Figure 5: **Ablating base vision encoder, swapping out CLIP ViT-L/14 from LLaVA for DFN-CLIP ViT-H/14.** We find that switching to DFN-CLIP improves fine-grained performance by +4.5pp and general performance by +1.2pp, given enough pretraining.

4 ABLATIONS: BASE MODELS AND TRAINING

4.1 EXPERIMENTAL SETUP

Building on our observational results in Section 3, we ablate key differences between models that may contribute to fine-grained classification performance, focusing primarily on base model selection and training methodology. We build on the LLaVA-1.5 training framework and alignment architecture (Liu et al., 2023). All models undergo one epoch of pretraining (when applicable) followed by one epoch of finetuning. We use LLaVA-1.5’s default hyperparameters, including learning rate and batch size, across all experiments.

Fine-grained performance is evaluated as the average score across the four classification benchmarks described in Section 3.2. For general VQA performance, we report the average score on three widely-used multiple-choice benchmarks: MMMU (Yue et al., 2024), MMBench (Liu et al., 2024b), and MMStar (Chen et al., 2024b). These benchmarks were selected for their domain coverage and widespread adoption in VLM evaluation.

4.2 BASE MODELS

Our analysis in Section 3 indicates that recent models typically incorporate newer base LLMs, though most continue to use the CLIP ViT-L/14 vision encoder from Radford et al. (2021). We investigate how these base model choices specifically impact fine-grained classification performance.

4.2.1 LLM CHOICE

Setup. Using LLaVA-1.5-7B (Liu et al., 2023) as our baseline, we ablate the LLM component across different vision encoders and pretraining configurations. We compare the original Vicuna-7B-1.5 (Chiang et al., 2023) (used in LLaVA) against Qwen2-7B (Yang et al., 2024) (used in Qwen2-VL-7B (Wang et al., 2024b) and Molmo-7B-D (Deitke et al., 2024)), as well as Llama-2-7B (Touvron et al., 2023) (Vicuna’s base model) and Qwen2-7B-Instruct (the instruction-tuned variant of Qwen2-7B).

Findings. As shown in Figure 4, replacing Vicuna-7B with Qwen2-7B substantially improves performance across all configurations, with average gains of +7.5pp (percentage points) on fine-grained benchmarks and +7.4pp on general VQA benchmarks. The instruction-tuned Qwen2-7B-Instruct shows similar improvements (+7.5pp and +8.1pp, respectively). Llama-2-7B, the non-instruction-tuned base of Vicuna, yields modest improvements, though the downstream implications of using a model without instruction tuning remain unclear. Notably, all our trained models—including the lower-performing Vicuna-based VLMs—consistently produce multiple-choice responses in the cor-

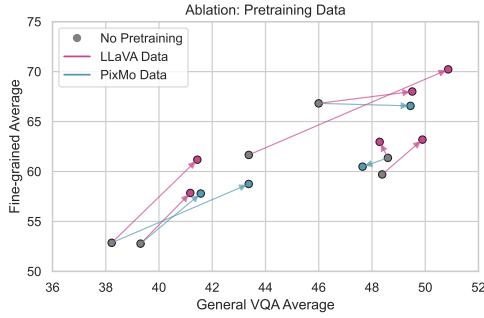


Figure 6: **Ablating pretraining data.** Adding either LLaVA (CC-3M) or Molmo (PixMo-Cap) data gives similar gains. LLaVA data increases fine-grained and general scores by +4.0pp and +2.1pp respectively, while PixMo data increases them by +2.4pp and +2.5pp.

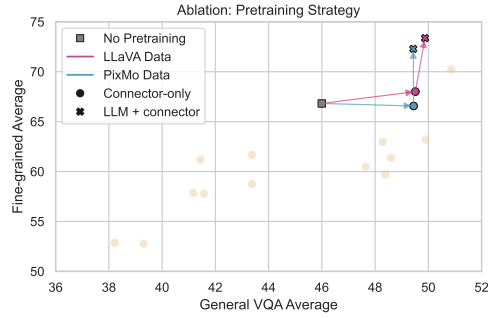


Figure 7: **Ablating training method, comparing connector-only pretraining with LLM+connector tuning.** Unfreezing the LLM during pretraining yields a +5.5pp improvement in fine-grained performance without adversely affecting general VQA scores.

rect format. This suggests that performance differences in fine-grained classification stem from improved knowledge about the presented options rather than better format adherence.

Takeaway 1: Stronger language models consistently improve performance across both fine-grained classification and general VQA benchmarks.

4.2.2 VISION ENCODER CHOICE

Setup. Most evaluated VLMs, including the LLaVA series and Molmo, use OpenAI-CLIP ViT-L/14-336px (Radford et al., 2021) as their vision encoder. However, Qwen2-VL (Wang et al., 2024b) uses DFN-CLIP ViT-H/14-378px (Fang et al., 2023), which demonstrates superior zero-shot performance on object recognition tasks like ImageNet. We evaluate both vision encoders across various LLM choices and pretraining configurations.

Findings. Figure 5 illustrates that the impact of substituting CLIP L/14 with DFN-CLIP H/14 depends on whether connector pretraining is performed before finetuning. Without pretraining, DFN-CLIP degrades general VQA performance (-2.5pp) while modestly improving fine-grained benchmark scores (+2.8pp). However, on models that have undergone connector pretraining, general VQA performance remains stable or slightly improves (+1.2pp) while fine-grained performance increases substantially (+4.5pp). This indicates that enhanced vision encoders can significantly boost fine-grained classification performance in pretrained VLMs.

Takeaway 2: Better vision encoders (like DFN-CLIP vs. CLIP) in VLMs improve fine-grained classification, but only when these encoders are properly integrated through pretraining before the finetuning stage.

4.3 TRAINING METHODS

Recent VLMs are typically trained in two stages (Liu et al., 2023; Wang et al., 2024b; Chen et al., 2024c). The first stage, pretraining, involves training on large-scale image-caption datasets using a captioning task to establish alignment between the vision encoder and LLM. The second stage, instruction tuning, refines the VLM with carefully curated multi-round, multi-modal instruction completion data, aligning it with human intent for real-world applications. Here, we explore the impact of these two training stages on the fine-grained capabilities of VLMs.

4.3.1 PRETRAINING

Setup. We investigate several critical questions about the pretraining process:

Table 1: **Summary of important ablation settings and comparison to existing models (bolded).** Upgrading the base LLM and vision encoder, as well as pretraining the connector and LLM on sufficient data, provide benefits to fine-grained performance, while instruction finetuning has a smaller effect. This is in contrast to general VQA benchmarks, where choice of LLM plays a larger part.

Model/Ablation	Vision Encoder	LLM	Arch.	Pretraining Data	Finetuning Data	Fine-grained Classification	General VQA
LLaVA-1.5-7B	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	59.3	41.8
No pretraining	CLIP L/14	Vicuna	LLaVA	None	LLaVA	52.8	39.3
LLaVA reproduction	CLIP L/14	Vicuna	LLaVA	LLaVA	LLaVA	57.9 (+5.1)	41.2 (+1.9)
Qwen2 LLM	CLIP L/14	Qwen2	LLaVA	LLaVA	LLaVA	63.0 (+5.1)	48.3 (+7.1)
DFN-CLIP encoder	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	68.0 (+5.0)	49.5 (+1.2)
Unfreeze LLM	DFN H/14	Qwen2	LLaVA	LLaVA	LLaVA	73.4 (+5.4)	49.9 (+0.4)
Pretrain on PixMo	CLIP L/14	Qwen2	LLaVA	PixMo	LLaVA	66.6	49.4
Molmo-7B-D	CLIP L/14	Qwen2	Molmo	PixMo	Molmo	68.4	58.0
FT Qwen2-VL base on LLaVA	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	LLaVA	85.5	61.1
Qwen2-VL-7B	DFN H/14	Qwen2	Qwen2-VL	Qwen2-VL	Qwen2-VL	87.9 (+2.4)	62.4 (+1.3)

- Is pretraining necessary for LLaVA-architecture models when the base models are already pretrained on billion-scale data?
- How does connector-only pretraining compare with unfreezing the LLM?
- How important is the quality of pretraining data for downstream performance?

We experiment with three pretraining configurations: LLaVA pretraining data (Liu et al., 2023) (a subset of CC-3M with web-scraped captions), Molmo pretraining data (Deitke et al., 2024) (PixMo-Cap, with high-quality human-annotated detailed captions), and no pretraining. For most experiments, we pretrain only the connector following Liu et al. (2023). Additionally, we explore a strategy similar to Deitke et al. (2024), where we pretrain the connector for 20% of the steps before unfreezing the LLM for the remaining 80%. All configurations are followed by one epoch of finetuning on LLaVA finetuning data.

Findings. Figure 6 illustrates our pretraining ablation results. We observe that connector-only pretraining generally enhances VLM performance, with the exception of the Qwen2 and CLIP ViT-L/14 combination. The benefits are particularly substantial for Vicuna-based VLMs, which show an average improvement of +6.1pp in fine-grained classification and +3.1pp in general VQA performance.

Takeaway 3: Large-scale pretraining on image captioning datasets substantially improves fine-grained classification performance but has a more modest effect on general VLM benchmarks.

Surprisingly, we find minimal differences between pretraining on low-quality web-scraped captions (LLaVA) versus highly detailed human annotations (PixMo). Compared to LLaVA data, PixMo data results in a -1.6pp change in fine-grained performance and a +0.4pp change in general VQA scores.

Takeaway 4: Pretraining data quality has a limited impact on overall model performance.

We hypothesize that connector-only pretraining may prevent the model from fully leveraging higher-quality captions. To test this hypothesis, we experiment with pretraining both the LLM and connector, incorporating a 20% connector-only warmup phase. Figure 7 shows that this approach significantly enhances fine-grained classification performance (+5.5pp) without compromising general VQA scores. However, this improvement occurs with both LLaVA and PixMo pretraining data, suggesting that training on the lower-quality captions does not impair the LLM’s language capabilities.

Takeaway 5: Pretraining both LLM and connector substantially enhances fine-grained benchmark performance while maintaining general VQA scores.

4.3.2 FINETUNING

Setup. Departing from our previous setup, we begin with the pretrained Qwen2-VL-7B-Base model (Wang et al., 2024b) (from which Qwen2-VL-7B-Chat is trained) and finetune on LLaVA-Instruct data (Liu et al., 2023) following our earlier experimental protocol. We compare these results

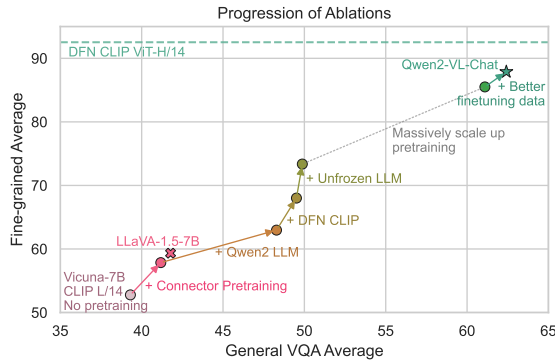


Figure 8: **Progression of combined ablation effects.** Starting with Vicuna-7B and CLIP ViT-L/14, we progressively modify the base LLM, vision encoder, and training settings to increase fine-grained classification accuracy from 52.8% to 73.4%, accounting for more than half the gap to Qwen2-VL-Chat’s 87.9%. We attribute the remaining performance difference primarily to Qwen2-VL’s extensive pretraining data of 1.4T tokens.

with Qwen2-VL-7B-Chat, which employs 2M in-house examples for instruction finetuning (Wang et al., 2024b). This comparison helps isolate the effects of different finetuning datasets.

Findings. Our results indicate that finetuning solely on LLaVA finetuning data slightly reduces both fine-grained and general benchmark performance. Specifically, fine-grained scores decrease by 2.4pp (from 87.9% to 85.5%), while general benchmark scores drop by 1.3pp (from 62.4% to 61.1%). As illustrated in Figure 8, this represents the smallest contribution to fine-grained performance across all our ablations, suggesting that instruction finetuning plays a less critical role in fine-grained knowledge acquisition compared to base model selection and pretraining strategy.

Takeaway 6: The instruction finetuning stage has comparatively less impact on fine-grained classification performance than other factors.

4.4 SUMMARY OF ABLATIONS

Figure 8 and Table 1 synthesize our ablation studies, showing how each component contributes to closing the performance gap between the weakest and strongest VLMs on both fine-grained and general VQA tasks. Beginning with the baseline LLaVA architecture (Liu et al., 2023) (Vicuna-7B (Chiang et al., 2023) and CLIP ViT-L/14 (Radford et al., 2021)), we systematically modified various components to increase fine-grained performance from 52.8% to 73.4%, and general VQA performance from 39.3% to 49.9%. This analysis reveals that switching to DFN-CLIP (Fang et al., 2023) and unfreezing the LLM during pretraining disproportionately enhances fine-grained capabilities compared to general VLM abilities, whereas switching the LLM to Qwen2-7B (Yang et al., 2024) accounts for most of the increase in general VQA performance.

Despite these substantial improvements, a notable 12-point gap in fine-grained classification performance remains unexplained by our ablations (Table 1). We consider two potential factors to account for this discrepancy: architectural differences and pretraining data scale.

While architectural changes could contribute to performance differences, our observational results (Section 3.4) and training ablations (Section 4.3) strongly suggest that pretraining data scale is the dominant factor. Our experiments with LLaVA (Liu et al., 2023) and PixMo (Deitke et al., 2024) data involved relatively small datasets—each comprising fewer than 1M images and captions, or approximately 200M and 400M tokens, respectively. In contrast, Wang et al. (2024b) report pretraining Qwen2-VL on an extensive 1.4T tokens—orders of magnitude more than in our experiments. This substantial disparity in pretraining data scale likely accounts for the remaining performance gap, highlighting the critical role of extensive pretraining in developing VLMs with superior fine-grained classification capabilities.

5 LIMITATIONS & CONCLUSION

Our study has some practical limitations that open up opportunities for future research. Due to computational constraints, we could only compare training on $<1\text{M}$ data points rather than the billion (B) scale training utilized by some newer VLMs, leaving open the question of how large-scale training might impact our observations. Additionally, newer work suggest different pretraining strategies, which might interact differently with fine-grained visual understanding.

In this work, we systematically evaluate state-of-the-art vision-language models (VLMs) on fine-grained classification benchmarks, highlighting fine-grained visual classification as a crucial yet underexplored dimension of VLMs. Through an in-depth analysis of key model components and training paradigms, we provide insights into strategies for improving fine-grained classification and enhancing vision-centric capabilities, ultimately strengthening the applicability of VLMs in real-world scenarios that demand precise visual understanding.

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Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	Classification	VLM
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.3	41.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	59.0	41.2
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.8	39.3
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	52.5	41.3
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	56.2	40.0
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	52.9	38.2
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.4	48.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	66.8	46.0
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	59.7	48.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	61.7	43.4
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	85.5	61.1
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.9	41.2
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	61.2	41.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	63.0	48.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	68.0	49.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	63.2	49.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	70.2	50.9
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	87.9	62.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	41.6
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	58.8	43.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	60.5	47.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	66.6	49.4
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	73.4	49.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	72.3	49.4

Table 2: Detailed results of model ablations.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	MMBench	MMMU	MMStar
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	32.2	32.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	58.6	32.9	32.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	54.4	32.8	30.7
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	58.0	34.3	31.7
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	53.9	34.6	31.5
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	51.3	33.2	30.1
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	63.1	41.8	40.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	61.3	36.2	40.5
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	63.2	41.4	40.5
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	55.7	38.3	36.1
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	78.5	49.0	55.8
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	60.4	29.9	33.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	57.4	34.2	32.7
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.4	41.3	39.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	64.9	41.3	42.3
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	66.5	41.3	41.9
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	68.0	41.6	43.0
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	78.9	49.9	58.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	57.8	32.6	34.3
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	61.1	34.1	34.9
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	64.9	40.0	38.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	63.0	41.0	44.3
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	66.0	42.0	41.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	63.7	40.0	44.6

Table 3: Detailed results of model ablations on general VQA benchmarks.

A FULL RESULTS

We present the complete ablation results for aggregated performance, general VQA benchmarks, and fine-grained classification in Table 2, Table 3, and Table 4, respectively.

Vision Encoder	LLM	Arch	PT Data	FT Data	Updating	ImageNet	Flowers	Pets	Food
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	70.9	50.7	46.3	69.4
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	68.7	50.4	46.5	70.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	None	LLaVA	Connector	64.3	46.1	40.4	60.4
CLIP ViT-L/14@336	Llama2-7B	LLaVA	None	LLaVA	Connector	64.4	44.2	40.7	60.8
DFN ViT-H/14@378	Llama2-7B	LLaVA	None	LLaVA	Connector	65.4	50.5	46.2	62.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	None	LLaVA	Connector	62.3	48.7	39.2	61.3
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	None	LLaVA	Connector	70.6	44.1	55.6	75.2
DFN ViT-H/14@378	Qwen2-7B	LLaVA	None	LLaVA	Connector	74.1	50.8	62.3	80.1
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	71.4	39.6	53.1	74.7
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	None	LLaVA	Connector	72.0	46.0	54.1	74.5
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	LLaVA	Full	84.8	79.5	87.1	90.7
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	69.1	50.0	45.3	67.1
DFN ViT-H/14@378	Vicuna-7B	LLaVA	LLaVA	LLaVA	Connector	72.7	53.2	48.5	70.4
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	71.5	46.9	56.6	76.9
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Connector	76.1	54.1	59.1	82.7
CLIP ViT-L/14@336	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	74.5	45.8	55.1	77.4
DFN ViT-H/14@378	Qwen2-7B-Instruct	LLaVA	LLaVA	LLaVA	Connector	78.5	52.3	65.4	84.7
DFN ViT-H/14@378	Qwen2-7B	Qwen2-VL	Qwen2-VL	Qwen2-VL	Full	85.9	82.3	91.0	92.3
CLIP ViT-L/14@336	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	69.2	47.4	46.7	67.8
DFN ViT-H/14@378	Vicuna-7B	LLaVA	PixMo	LLaVA	Connector	70.8	50.9	44.8	68.5
CLIP ViT-L/14@336	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	70.8	43.7	53.9	73.6
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Connector	76.1	50.3	57.8	82.1
DFN ViT-H/14@378	Qwen2-7B	LLaVA	LLaVA	LLaVA	Full	79.4	60.7	67.3	86.0
DFN ViT-H/14@378	Qwen2-7B	LLaVA	PixMo	LLaVA	Full	80.0	56.4	65.4	87.4

Table 4: Detailed results of model ablations on fine-grained classification benchmarks.

B EVALUATION

B.1 FINE-GRAINED CLASSIFICATION

As covered in Section 3.2, we evaluate VLMs and base vision encoders on four different traditional classification benchmarks:

ImageNet-1K (Deng et al., 2009): A foundational image classification dataset covering a broad range of supercategories, each containing multiple fine-grained subcategories, based on the WordNet hierarchy. The ImageNet test set contains 50,000 images with classification across 1,000 classes.

However, because of the nature of the ImageNet/WordNet class hierarchy, there are many classes which are too visually distinct from other classes to create hard multiple choice examples for (for instance, there are only three classes under the “person” category). Due to this drawback, in addition to the excessive size of the dataset, we employ human annotated incorrect model predictions from Recht et al. (2019) to select images for which hard negative answer choices can be constructed. Starting with 19,056 annotated examples, we narrow down the pool to 5,457 images consisting of 928 different classes out of the original 1,000. Furthermore, since this test example curation leaves the remaining classes unevenly balanced, we report the mean-per-class accuracy for ImageNet.

Oxford Flowers-102 (Nilsback & Zisserman, 2008): A dataset of 102 flower species characterized by high intra-class variation and inter-class similarity. The Flowers dataset consists of 6,149 test examples.

Oxford-IIIT Pet-37 (Parkhi et al., 2012): A collection of 37 pet categories with challenging variations in pose, lighting, and occlusion. The Pets dataset consists of 3,669 test examples.

Food-101 (Bossard et al., 2014): A dataset comprising 101 food categories, exhibiting substantial visual diversity within each class. The Food dataset consists of 25,250 test examples.

Question generation. For all fine-grained classification test sets except the ImageNet dataset (described above), we generate hard negatives by performing zero-shot classification on the test examples using an OpenCLIP ViT-L/14 trained on LAION (Schuhmann et al., 2022), following Geigle et al. (2024). We take the top 4 predictions that are incorrect, and shuffle them with the correct label to produce the list of answer choices.

Prompt formatting. In our multiple choice VQA format, the question is always of the form “What type of object is in this photo?” The answer choices are formatted as “A. Option 1\n B. Option 2\n C. Option 3\n D. Option 4\n E. Option 5”. However, since different VLMs are trained with different multiple choice prompts and format, we use each model’s corresponding format from VLMEvalKit (Duan et al., 2024), for instance: “Answer with the option’s letter from the given choices directly.”

For CLIP models, we use the standard zero-shot evaluation procedure, and thus there is no question format. The classnames passed to the text encoder are instead formatted in the template of “a photo of a {classname}”.

B.2 GENERAL VQA

In our initial observational testing from Section 3, we rely directly on the numbers reported on the OpenVLM Leaderboard (Duan et al., 2024). The default leaderboard reports the average score across eight diverse VLM benchmarks: MMBench (Liu et al., 2024b), MMStar (Chen et al., 2024b), MMMU (Val) (Yue et al., 2024), MathVista (Lu et al., 2024), OCRBench (Liu et al., 2024c), AI2D (Kembhavi et al., 2016), HallusionBench (Guan et al., 2024), and MMVet (Yu et al., 2024). This reported average score is used only in Figure 2 in our paper.

For all further evaluations, both on existing models and our own models trained in Section 4, we narrow this down to a subset of multiple-choice VLM benchmarks for a more direct comparison of performance. Specifically, we evaluate VLMs on three benchmarks:

MMBench (Liu et al., 2024b): A dataset of 2,948 examples testing a wide range of VLM capabilities split between perception and reasoning. While the majority of the examples deal with coarse-grained perception and visual reasoning tasks, some examples test “fine-grained perception”, though the categories (such as action recognition, attribute recognition, and OCR) are in a different domain from the fine-grained classification datasets we consider.

MMMU (Val) (Yue et al., 2024): A large-scale multimodal dataset covering a wide distribution of domains and applications. Though the test set has 10,500 examples, most works cite model performance on the validation split, which contains 900 questions. MMMU tests general knowledge tied to visual perception across art, science, engineering, medicine, business, and humanities.

MMStar (Chen et al., 2024b): A dataset comprised of 1,500 questions selected by human annotators. The examples are chosen to cover a diverse set of tasks, both across domain and low-level model capabilities.

Evaluation procedure. For the general VQA benchmarks, we defer entirely to the VLMEvalKit evaluation code (Duan et al., 2024). All existing models have a pre-defined evaluation procedure, and for our own trained models, we apply the LLaVA evaluation code, since we exclusively finetune on LLaVA-Instruct data.

C TRAINING

We build on the LLaVA-1.5 training codebase (Liu et al., 2023). For the majority of experiments, we stick to the LLaVA architecture and training procedure.

Pretraining. This means one epoch of pretraining on the LLaVA pretraining data, obtained from CC-3M images and captions, or Molmo pretraining data (PixMo-Cap). For PixMo-Cap, we only use the “caption” field and not the raw audio transcript. During pretraining, we use a batch size of 256. In most of our pretraining runs, we tune only the randomly-initialized MLP connector with a learning rate of $1e-3$. When ablating pretraining strategy, we first tune the connector only as warmup, for 20% of steps, then unfreeze the LLM and tune it as well, with a learning rate of $2e-5$ for the remaining 80% of steps.

Finetuning. For finetuning, we always train for one epoch on the LLaVA-1.5 finetuning data, which consists of LLaVA-Instruct and VQA training data from various datasets to allow proper benchmarking. During finetuning, we maintain a learning rate of $2e-5$ for the LLM, with a batch size of 128. The vision encoder remains frozen during both phases of training.

For the ablation experiment on the Qwen2-VL architecture, we start with Qwen2-VL-7B Base and train with the same method, unfreezing only the LLM.

Compute. We train all models using 4 nodes of 4 A100 (40GB) GPUs each. Training typically completes within 20 hours, with variation depending on the CLIP model size.

LLaVA vs. PixMo data. In our main text, we refer to the difference in caption quality between LLaVA pretraining data and PixMo pretraining data. LLaVA pretraining data is sourced from CC-3M, which consists of web-scraped captions, commonly known for low quality and relevance. PixMo-Cap is, on the other hand, collected from human annotators recording a 60–90 second audio description of an image, and then combined into a proper image caption (Deitke et al., 2024). A surface analysis of the two datasets shows that the LLaVA data averages 9.8 words per caption, whereas the PixMo data averages 169 words per caption.

LLaVA caption example: “front panel bracket cover for suzuki”

PixMo caption example: “The poster from the TV show ”Law and Order” features a past cast ensemble set in an interrogation room. On the left is a Hispanic man, identified as Detective Nick Amaro, portrayed by Danny Pino. He has dark hair, olive skin, and is dressed in a gray jacket, blue shirt, and black tie. Standing beside him is a blonde woman with shoulder-length hair, who is identified as Detective Amanda Rollins, played by Kelli Giddish. She is ... The backdrop is an interrogation room, with a concrete wall and a portion of a two-way mirror visible, adding to the procedural drama’s atmospheric setting.”